

Lab 3 - Routing Protocol

NAME: **WOO CHENG SHUAN**

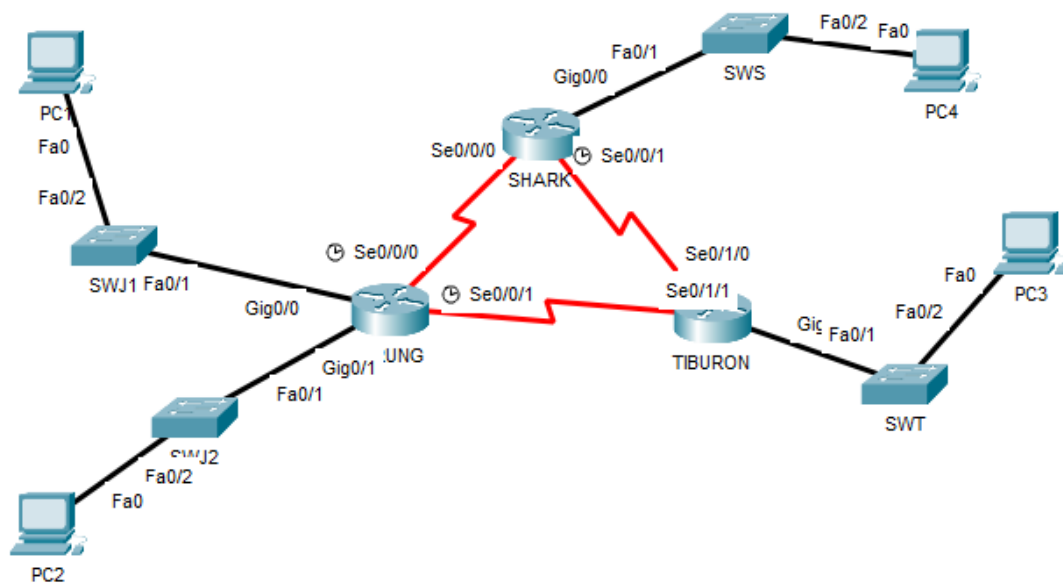
MATRIC NYM.: **A23CS0283**

SECTION: **12**

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.129	255.255.255.224	-
4		G0/1	172.18.110.161	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.65	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.1	255.255.255.192	-
11	PC1	-	172.18.110.158	255.255.255.224	172.18.110.129
12	PC2	-	172.18.110.190	255.255.255.224	172.18.110.161
13	PC3	-	172.18.110.126	255.255.255.192	172.18.110.65
14	PC4	-	172.18.110.62	255.255.255.192	172.18.110.1

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

[illegible]

172, 18, 110, 0 / 23

17 > .18. 0 1 1 0 1 1 1 0 . 0 0 0 0 0 0 0 0

8 + 8 + 7 = 23

borrow 6 bits - SCAN

borrow 6 bits - SCAN ← borrow 6 bits →
 172.18.01101110.0000000000000000

172. 18. 110. 0 / 26

0	0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1

NA: 172.16.110.0

BA: 172. 48. 110. 63

usable host : 172.18.110.1 - 172.18.110.62

borrow 6 bits — TLAN

borrow 6 bits — TLN
 ← borrow 6 bits →

172. 18. 0 | 1 0 1 1 0 . 0 0 0 0 0 0 0 0

172. 18. 110. 0 / 26

0	.	0	1	0	0	0	0	0	0
0	.	0	1	1	1	1	1	1	1

NA: 172 · 18 · 110 · 64

BA : 172 . 18 . 110 . 127

usable host: 172.18.110.65 — 172.18.110.126

borrow 5 bits — JUAN I

borrow 5 bits — JUAN!

172, 18. 0 1 1 0 1 1 1 0 . 0 0 0 0 0 0 0 0 0 0

← borrow 5 bits

172 . 18 . 110 . 0 / 27

0	1	0	0	0	0	0	0	0
0	1	0	0	1	1	1	1	1

NA: 172, 18, 110, 128

BA : 172. 18. 110. 159

usable host : 172.18.110.129 - 172.18.110.158

borrow 5 bits — XAN 2

borrow 5 bits — JCAN2
 H2, 18: 0 1 1 0 1 1 1 0 . 0 0 0 0 0 0 0 0

172. 18. 110. 0 / 27

0	1	0	1	0	0	0	0	0
0	1	0	1	1	1	1	1	1

NA: 172-18-110-160

BA: 172.18.110.191

usable host : 172.18.110.161 — 172.18.110.190

borrow 2 bits — J to S		borrow 2 bits	
172.18.0	11011100	00000000	00
172.18.110.0/30			
0	11000000	00	
0	11000001	11	
NA: 172.18.110.192			
BA: 172.18.110.195			
usable host: 172.18.110.193 — 172.18.110.194			

borrow 2 bits — J to T		borrow 2 bits	
172.18.0	11011100	00000000	00
172.18.110.0/30			
0	11000001	00	
0	11000011	11	
NA: 172.18.110.196			
BA: 172.18.110.199			
usable host: 172.18.110.197 — 172.18.110.198			

borrow 2 bits — S to T		borrow 2 bits	
172.18.0	11011100	00000000	00
172.18.110.0/30			
0	11000100	00	
0	11000101	11	
NA: 172.18.110.200			
BA: 172.18.110.203			
usable host: 172.18.110.201 — 172.18.110.202			

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - a. CLI
 - i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

ANS:

```
JERUNG>enable
JERUNG#show IP route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.128/27 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.160/27 is directly connected, GigabitEthernet0/1
L       172.18.110.161/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#
```

```

TIBURON>enable
TIBURON#show IP route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#

SHARK>enable
SHARK#show IP route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

SHARK#

```

- b. PT tool
 - i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

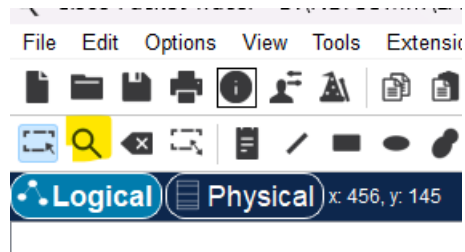


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

ANS:

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
L	172.18.110.161/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

Routing Table for TIBURON

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

2. Try to ping PC2 from PC1, paste the results here.

ANS: SUCCESS

```

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Reply from 172.18.110.190: bytes=32 time=14ms TTL=128
Reply from 172.18.110.190: bytes=32 time<1ms TTL=128
Reply from 172.18.110.190: bytes=32 time=6ms TTL=128
Reply from 172.18.110.190: bytes=32 time<1ms TTL=128

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 5ms

```

3. Try to ping PC4 from PC1, paste the results here.

ANS: FAIL

```

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

4. Explain the reason(s) behind the results.

ANS: Ping PC1 and PC2 is successful because they are in same subnet which is JERUNG. However ping PC1 and PC4 is fail because they are not in the same subnet.

5. What needs to be done to ensure all PCs can ping each other successfully?

ANS: We need to configure the routers with the proper routes to include all network segments to ensures that the routers know how to forward traffic in between the PCs.

Task 2 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- a. What can you say about the addresses used in the 'network' instructions in Figure E?

ANS: The addresses used in the 'network' instructions is the network address of router JERUNG. They also act as the router information of JERUNG.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

ANS:

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	R	172.18.110.64/26	Serial0/0/1	172.18.110.198	120/1
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/0	---	0/0
R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1	L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1	C	172.18.110.160/27	GigabitEthernet0/1	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	L	172.18.110.161/32	GigabitEthernet0/1	---	0/0

JERUNG#show IP route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks

C 172.18.110.128/27 is directly connected, GigabitEthernet0/0

L 172.18.110.129/32 is directly connected, GigabitEthernet0/0

C 172.18.110.160/27 is directly connected, GigabitEthernet0/1

L 172.18.110.161/32 is directly connected, GigabitEthernet0/1

C 172.18.110.192/30 is directly connected, Serial0/0/0

L 172.18.110.193/32 is directly connected, Serial0/0/0

C 172.18.110.196/30 is directly connected, Serial0/0/1

L 172.18.110.197/32 is directly connected, Serial0/0/1

JERUNG#

TIBURON#show IP route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

172.18.0.0/16 is variably subnetted, 9 subnets, 4 masks

C 172.18.110.64/26 is directly connected, GigabitEthernet0/0

L 172.18.110.65/32 is directly connected, GigabitEthernet0/0

R 172.18.110.128/27 [120/1] via 172.18.110.197, 00:00:20, Serial0/1/1

R 172.18.110.160/27 [120/1] via 172.18.110.197, 00:00:20, Serial0/1/1

R 172.18.110.192/30 [120/1] via 172.18.110.197, 00:00:20, Serial0/1/1

C 172.18.110.196/30 is directly connected, Serial0/1/1

L 172.18.110.198/32 is directly connected, Serial0/1/1

C 172.18.110.200/30 is directly connected, Serial0/1/0

L 172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#

- What are the changes seen in TIBURON?
ANS: RIP protocol generated new routes and are added to the routing table marked with a "R" for RIP.
- What are the Networks with type R in TIBURON and JERUNG?
ANS: The networks with type R in TIBURON and JERUNG are Routing Information Protocol (RIP). RIP is used to manage router information in a self-contained network such as a corporate LAN or a private WAN. They will automatically be updated in the routers' routing table.
- Ping PC3 from PC1. Was it successful?
ANS: SUCCESS

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=3ms TTL=126
Reply from 172.18.110.126: bytes=32 time=2ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

```

- d. Ping PC4 from PC1. Was it successful?

ANS: FAIL

```

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.129: Destination host unreachable.
Reply from 172.18.110.129: Destination host unreachable.
Request timed out.
Reply from 172.18.110.129: Destination host unreachable.

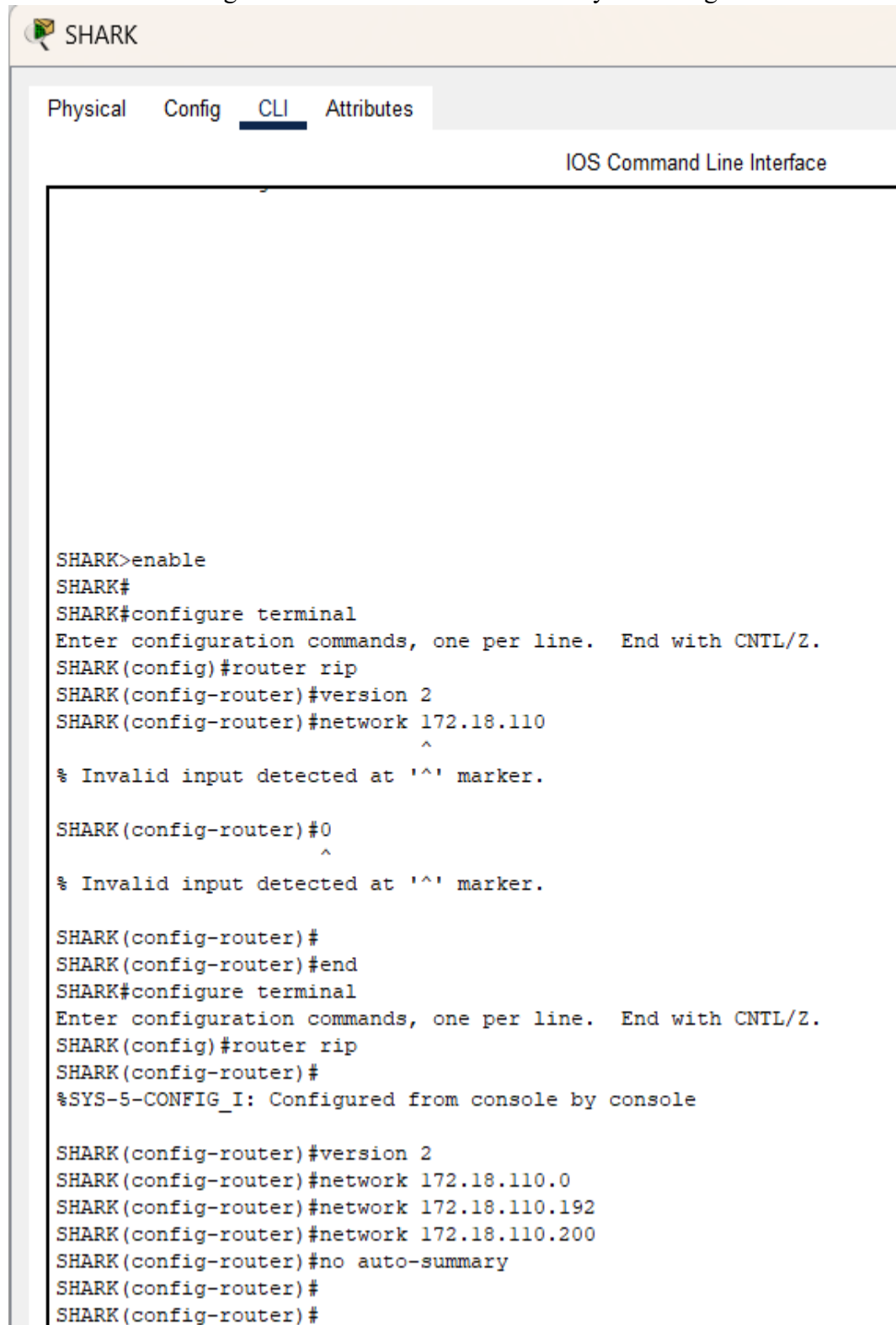
Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

- e. Explain the reasons for your answer.

ANS: The ping from PC1 to PC3 is successful because the network connection between them is correctly configured and the routes are set up to allow the communication. RIP routing protocol of JERUNG for PC1 and TIBURON for PC3 have properly setup and configured then automatically be updated in the routers' routing table. However, the ping from PC1 to PC4 is not successful because the network connection between them and routing table for SHARK are not properly configured.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.



The screenshot shows the SHARK router's CLI interface. The top bar has tabs for Physical, Config, CLI (selected), and Attributes. Below the tabs is the title 'IOS Command Line Interface'. The main area displays the following commands and their outputs:

```

SHARK>enable
SHARK#
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110
                                     ^
% Invalid input detected at '^' marker.

SHARK(config-router)#0
                                     ^
% Invalid input detected at '^' marker.

SHARK(config-router)#
SHARK(config-router)#end
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#
%SYS-5-CONFIG_I: Configured from console by console

SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.0
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
SHARK(config-router)#

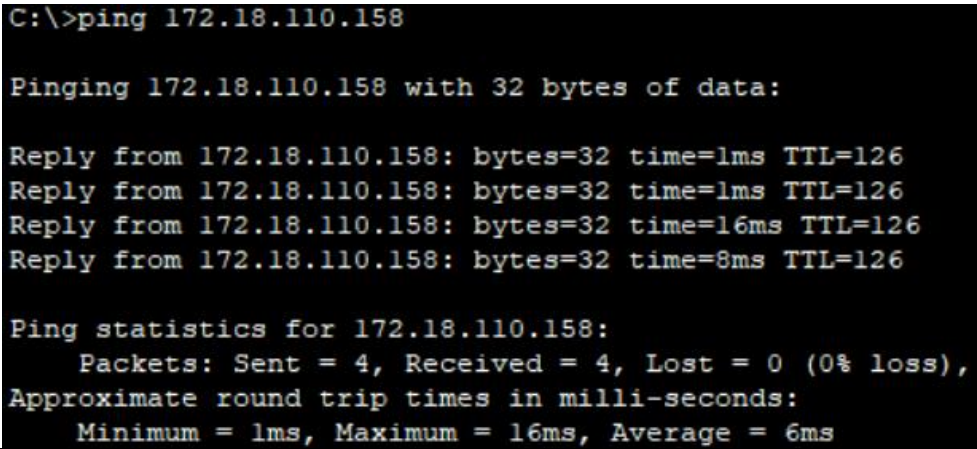
```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	SUCCESS  <pre> C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=16ms TTL=126 Reply from 172.18.110.158: bytes=32 time=8ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 16ms, Average = 6ms </pre>
2	SUCCESS

	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Request timed out. Reply from 172.18.110.190: bytes=32 time=4ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 4ms, Average = 2ms </pre>
3	SUCCESS <pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Request timed out. Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: [192.168.1.0/24](#)
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
[ANS: FAIL](#)


```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.202: Destination host unreachable.
Reply from 172.18.110.202: Destination host unreachable.
Request timed out.
Reply from 172.18.110.202: Destination host unreachable.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

ANS: FAIL

```
C:\>ping 172.18.110.158

Pinging 172.18.110.158 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.158:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

ANS:

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	R	172.18.110.0/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	R	172.18.110.128/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/0/1	172.18.110.202	120/16	R	172.18.110.160/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.128/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.160/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	L	172.18.110.202/32	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

 SHARK

Physical
Config
CLI
Attributes

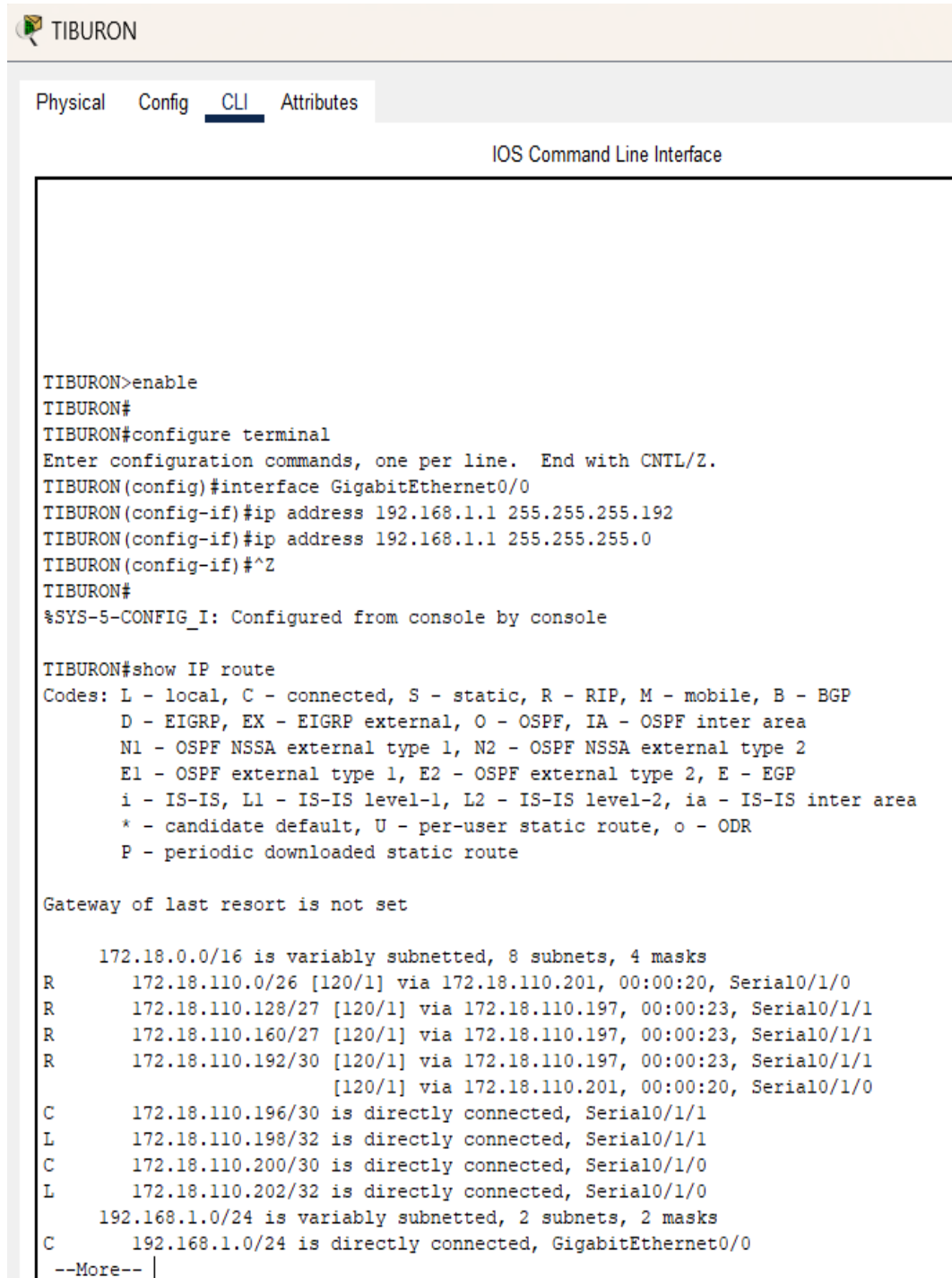
IOS Command Line Interface

Press RETURN to get started.

SHARK>enable
SHARK#show IP route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

172.18.0.0/16 is variably subnetted, 10 subnets, 4 masks
C 172.18.110.0/26 is directly connected, GigabitEthernet0/0
L 172.18.110.1/32 is directly connected, GigabitEthernet0/0
R 172.18.110.64/26 is possibly down, routing via 172.18.110.202, Serial0/0/1
R 172.18.110.128/27 [120/1] via 172.18.110.193, 00:00:12, Serial0/0/0
R 172.18.110.160/27 [120/1] via 172.18.110.193, 00:00:12, Serial0/0/0
C 172.18.110.192/30 is directly connected, Serial0/0/0
L 172.18.110.194/32 is directly connected, Serial0/0/0
R 172.18.110.196/30 [120/1] via 172.18.110.202, 00:00:25, Serial0/0/1
[120/1] via 172.18.110.193, 00:00:12, Serial0/0/0
C 172.18.110.200/30 is directly connected, Serial0/0/1
L 172.18.110.201/32 is directly connected, Serial0/0/1
--More-- |



The screenshot shows the TIBURON network simulator interface. At the top, there are tabs for 'Physical', 'Config', 'CLI' (which is selected), and 'Attributes'. Below the tabs, the title 'IOS Command Line Interface' is displayed. The main area shows a terminal session with the following commands and output:

```

TIBURON>enable
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#ip address 192.168.1.1 255.255.255.192
TIBURON(config-if)#ip address 192.168.1.1 255.255.255.0
TIBURON(config-if)#^Z
TIBURON#
%SYS-5-CONFIG_I: Configured from console by console

TIBURON#show IP route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 4 masks
R       172.18.110.0/26 [120/1] via 172.18.110.201, 00:00:20, Serial0/1/0
R       172.18.110.128/27 [120/1] via 172.18.110.197, 00:00:23, Serial0/1/1
R       172.18.110.160/27 [120/1] via 172.18.110.197, 00:00:23, Serial0/1/1
R       172.18.110.192/30 [120/1] via 172.18.110.197, 00:00:23, Serial0/1/1
           [120/1] via 172.18.110.201, 00:00:20, Serial0/1/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
--More-- |

```

- e. Referring to the routing table, explain your findings.

ANS: The new network address of the subnet has not updated into the routing table. Therefore, it causes all the packets that destined for PC3 to be dropped. Thus, PC4 and PC1 cannot ping PC3 after changing the new IP address.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

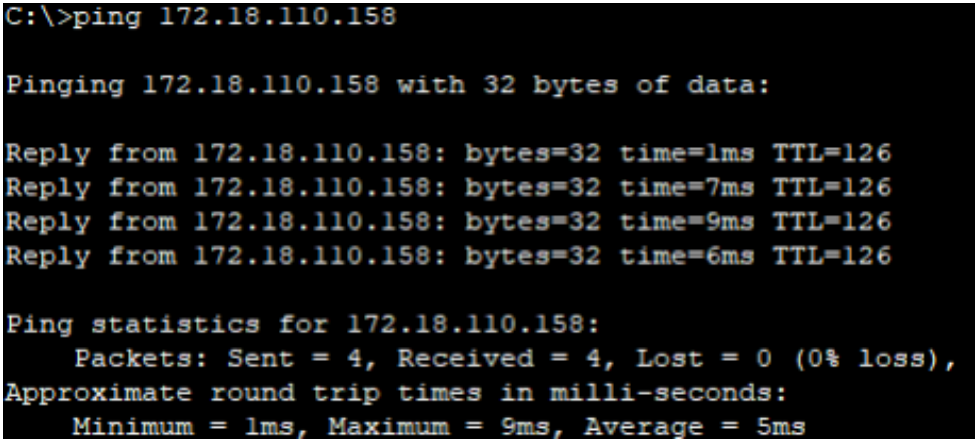
ANS: My next move to ensure end-to-end connectivity is to update the routing configurations in TIBURON to include the new subnet (192.168.1.0/24) and advertise it using the RIP protocol to ensure that it can be updated into the TIBURON routing table automatically.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

ANS:

```
TIBURON#
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#network 172.18.110.196
TIBURON(config-router)#network 172.18.110.200
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	SUCCESS  <pre>C:\>ping 172.18.110.158 Pinging 172.18.110.158 with 32 bytes of data: Reply from 172.18.110.158: bytes=32 time=1ms TTL=126 Reply from 172.18.110.158: bytes=32 time=7ms TTL=126 Reply from 172.18.110.158: bytes=32 time=9ms TTL=126 Reply from 172.18.110.158: bytes=32 time=6ms TTL=126 Ping statistics for 172.18.110.158: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 9ms, Average = 5ms</pre>
2	SUCCESS

	<pre> C:\>ping 172.18.110.190 Pinging 172.18.110.190 with 32 bytes of data: Reply from 172.18.110.190: bytes=32 time=9ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=1ms TTL=126 Reply from 172.18.110.190: bytes=32 time=12ms TTL=126 Ping statistics for 172.18.110.190: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 12ms, Average = 5ms </pre>
4	SUCCESS <pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms </pre>

REFLECTION

What have you learned in this task?

ANS: In this task, I learned the importance of subnetting, routing protocols and proper configuration for maintaining network connectivity. I also learned how to allocate IP addresses, configure routers using RIP and update routing tables to reflect network changes.

Troubleshooting failed connections highlighted the need for careful setup and verification. In a nutshell, the tasks emphasized adaptability and precision in managing and troubleshooting networks.